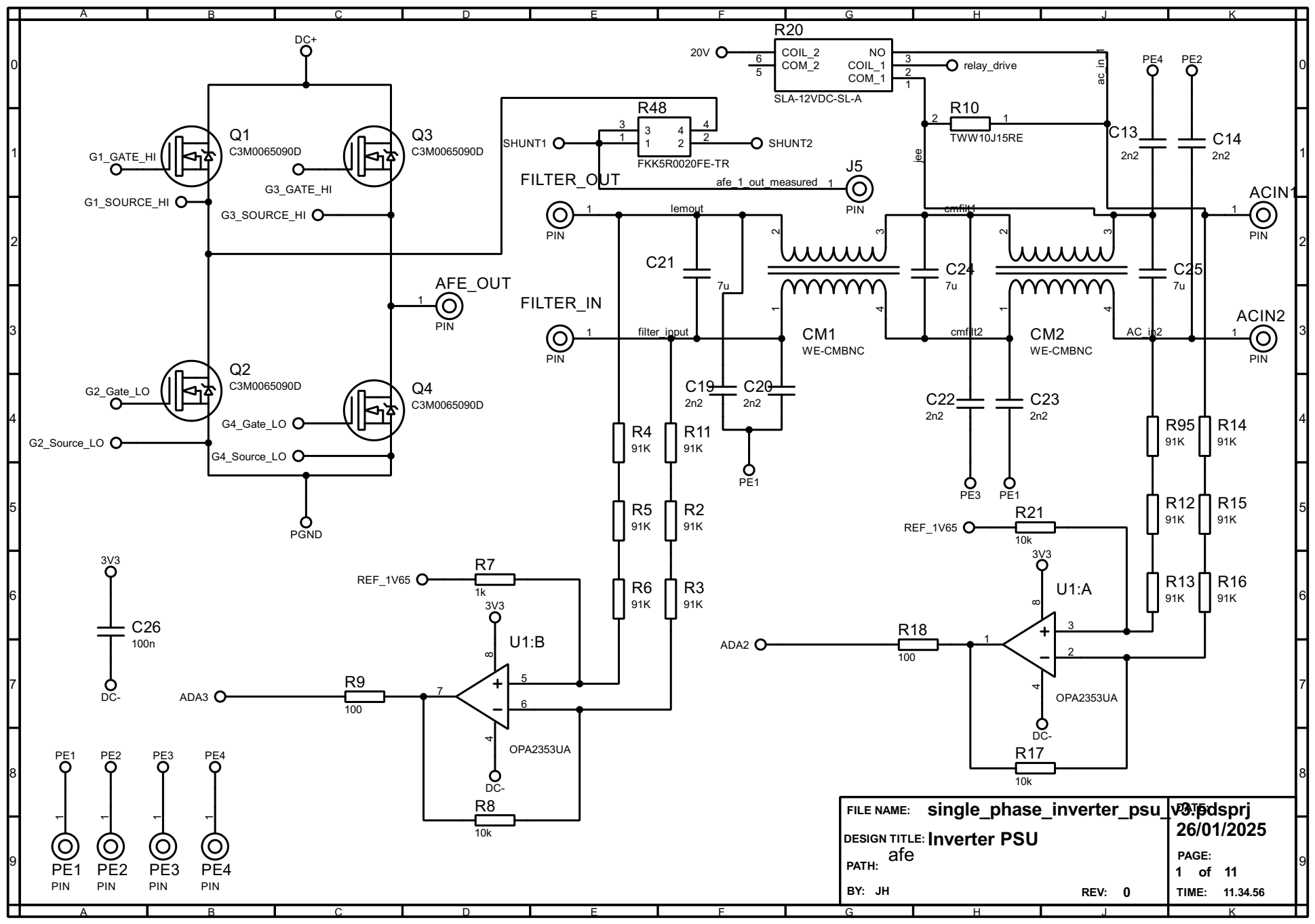
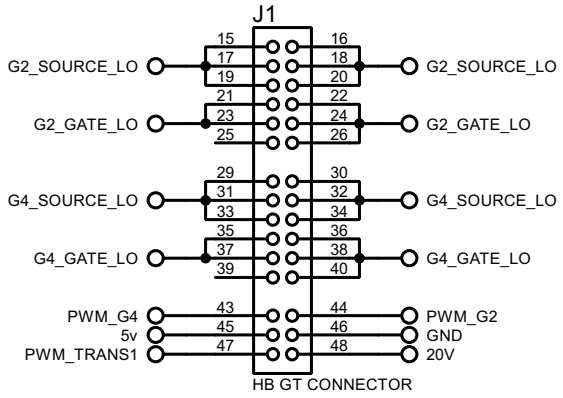
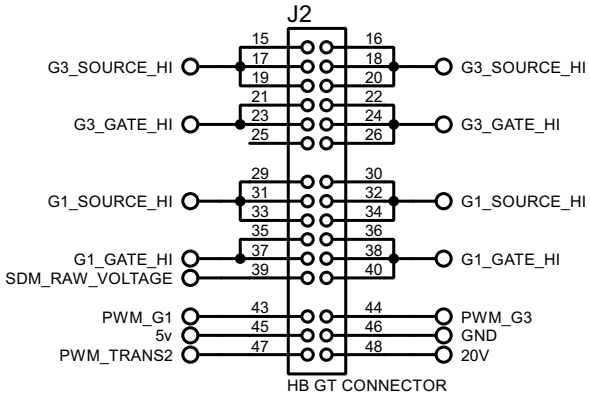
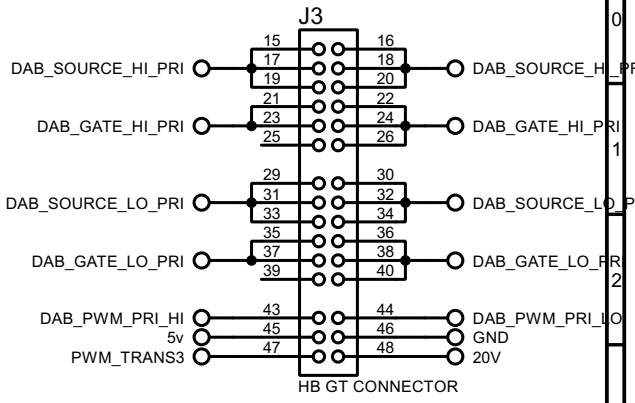
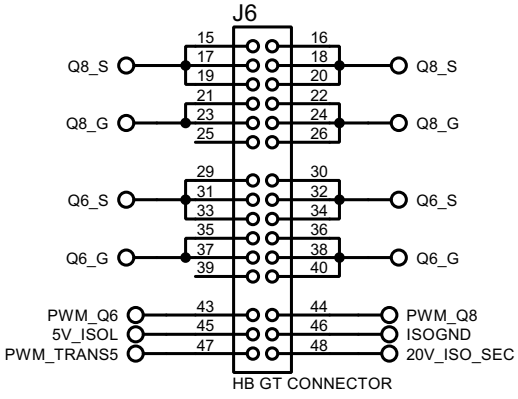
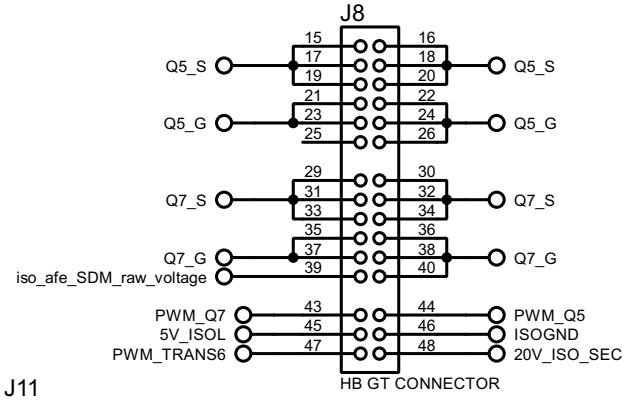

DC+
R20
20V
COIL_2
COM_2
NO
COIL_1
COM_1
relay_drive
SLA-12VDC-SL-A
R10
TWW10J15RE
PE4
PE2
C13
2n2
C14
2n2
R48
SHUNT1
SHUNT2
FKK5R0020FE-TR
FILTER_OUT
afe_1_out_measured
J5
PIN
ACIN1
Q1
C3M0065090D
G1_GATE_HI
G1_SOURCE_HI
Q3
C3M0065090D
G3_GATE_HI
G3_SOURCE_HI
AFE_OUT
lemout
cmfilt1
C21
7u
C24
7u
C25
7u
FILTER_IN
filter_input
CM1
WE-CMBNC
cmfilt2
CM2
WE-CMBNC
AC_in2
ACIN2
Q2
C3M0065090D
G2_Gate_LO
G2_Source_LO
Q4
C3M0065090D
G4_Gate_LO
G4_Source_LO
PGND
C19
2n2
C20
2n2
C22
2n2
C23
2n2
PE1
PE3
PE1
R4
91K
R11
91K
R5
91K
R2
91K
R6
91K
R3
91K
R95
91K
R14
91K
R12
91K
R15
91K
R13
91K
R16
91K
R21
10k
REF_1V65
3V3
U1:A
R18
100
ADA2
OPA2353UA
DC-
R17
10k
C26
100n
R7
1k
U1:B
R9
100
ADA3
R8
10k
PE1
PE2
PE3
PE4
FILE NAME: single_phase_inverter_psu_v3.pdsprj
DESIGN TITLE: Inverter PSU
afe
PATH:
BY: JH
REV: 0
DATE: 26/01/2025
PAGE: 1 of 11
TIME: 11.34.56

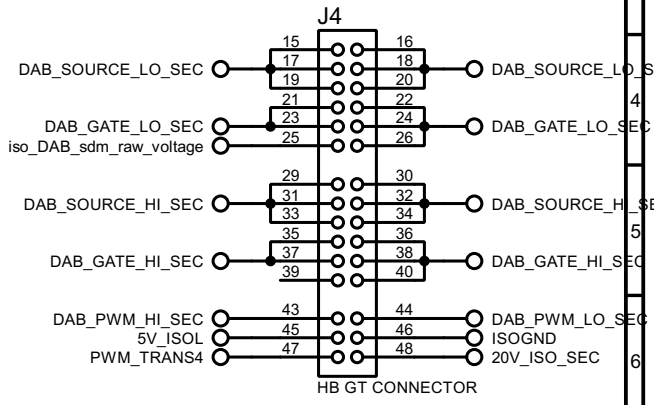


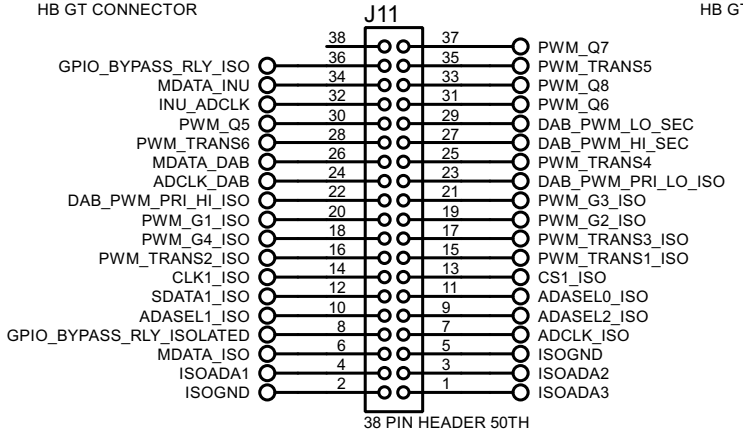


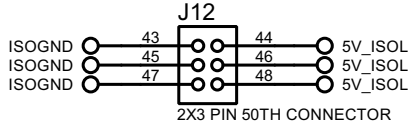




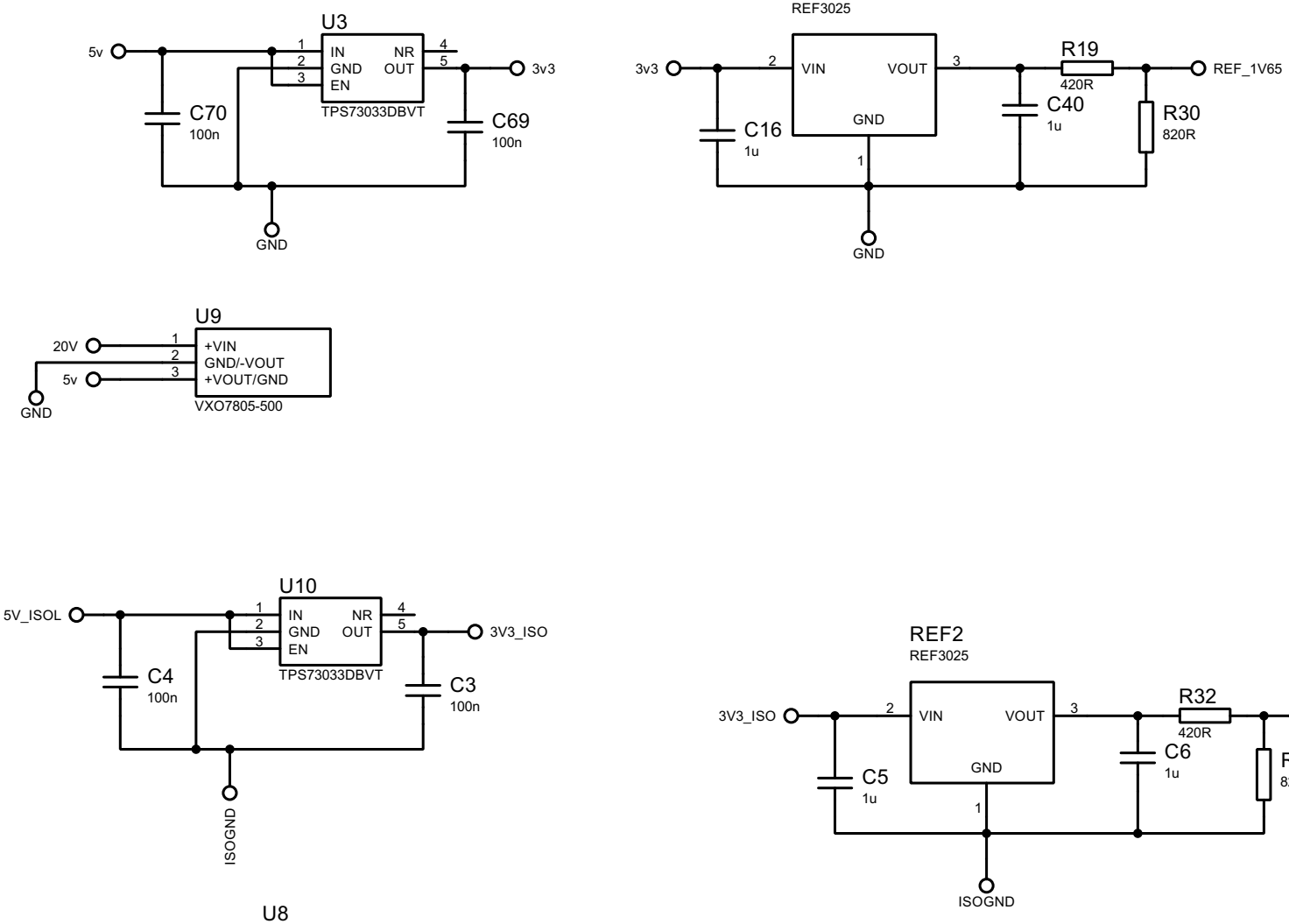




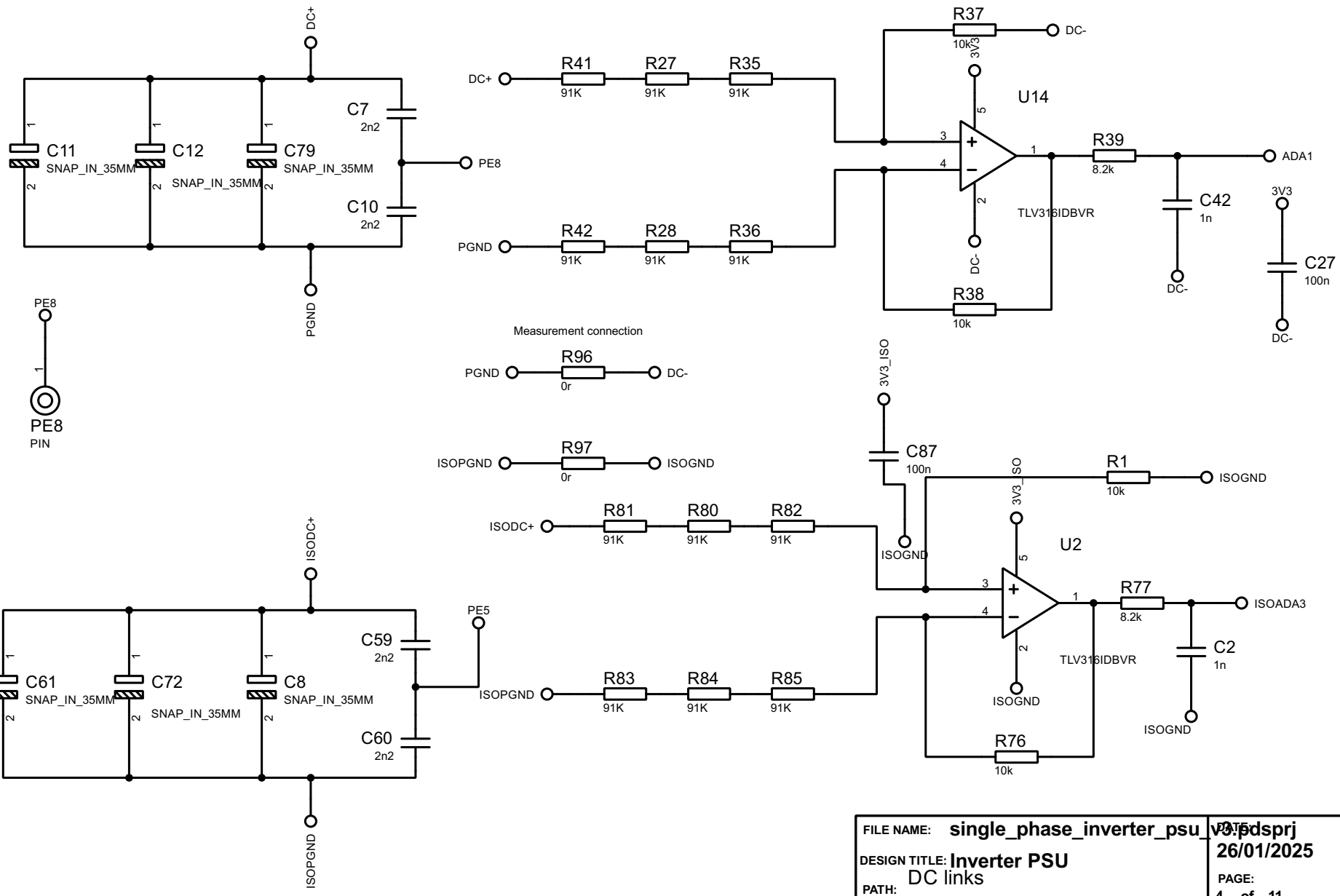




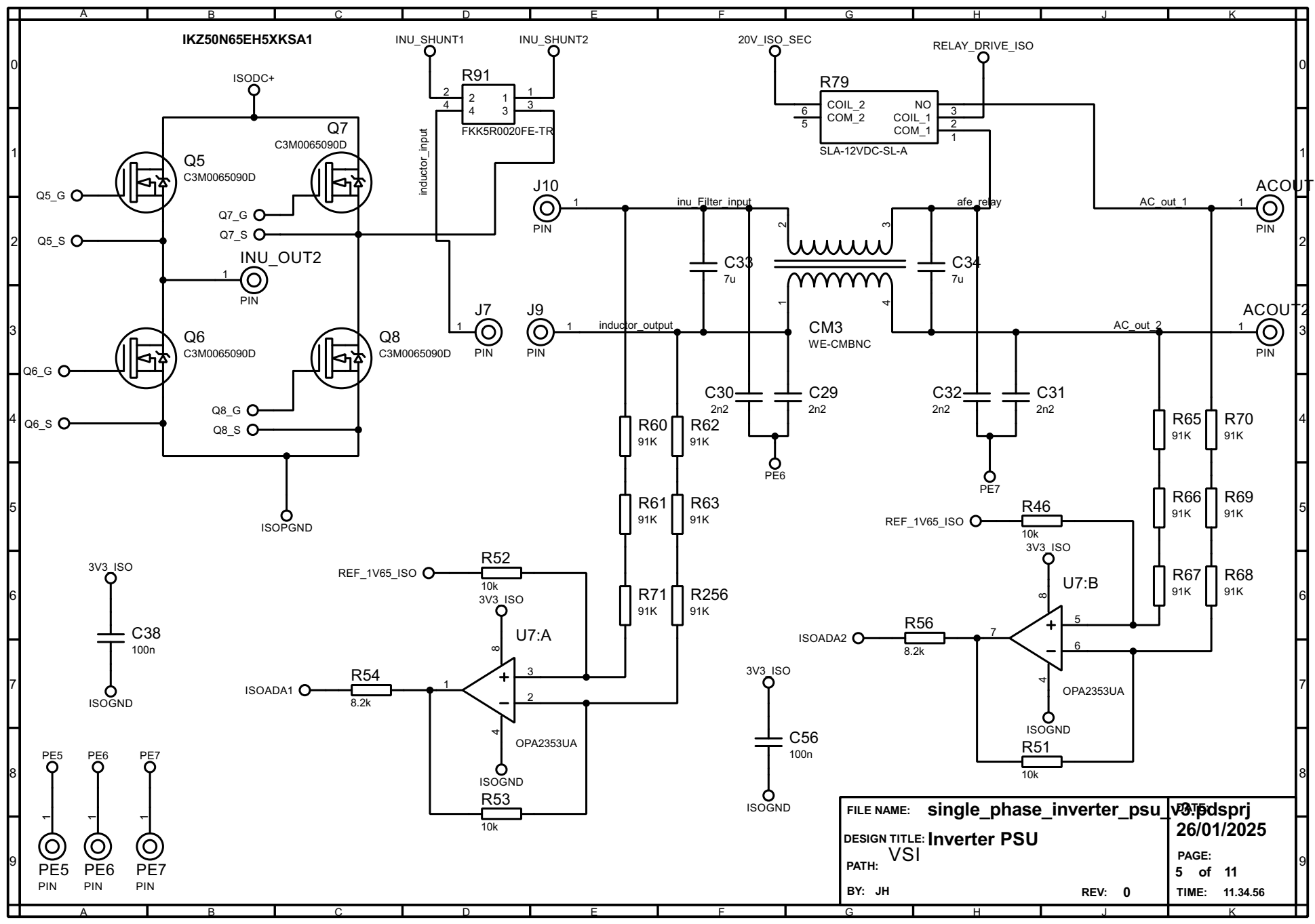
FILE NAME: single_phase_inverter_psu_v3.pdsprj	DATE: 26/01/2025
DESIGN TITLE: Inverter PSU fpga connector	PAGE: 2 of 11
PATH:	TIME: 11.34.56
BY: JH	REV: 0

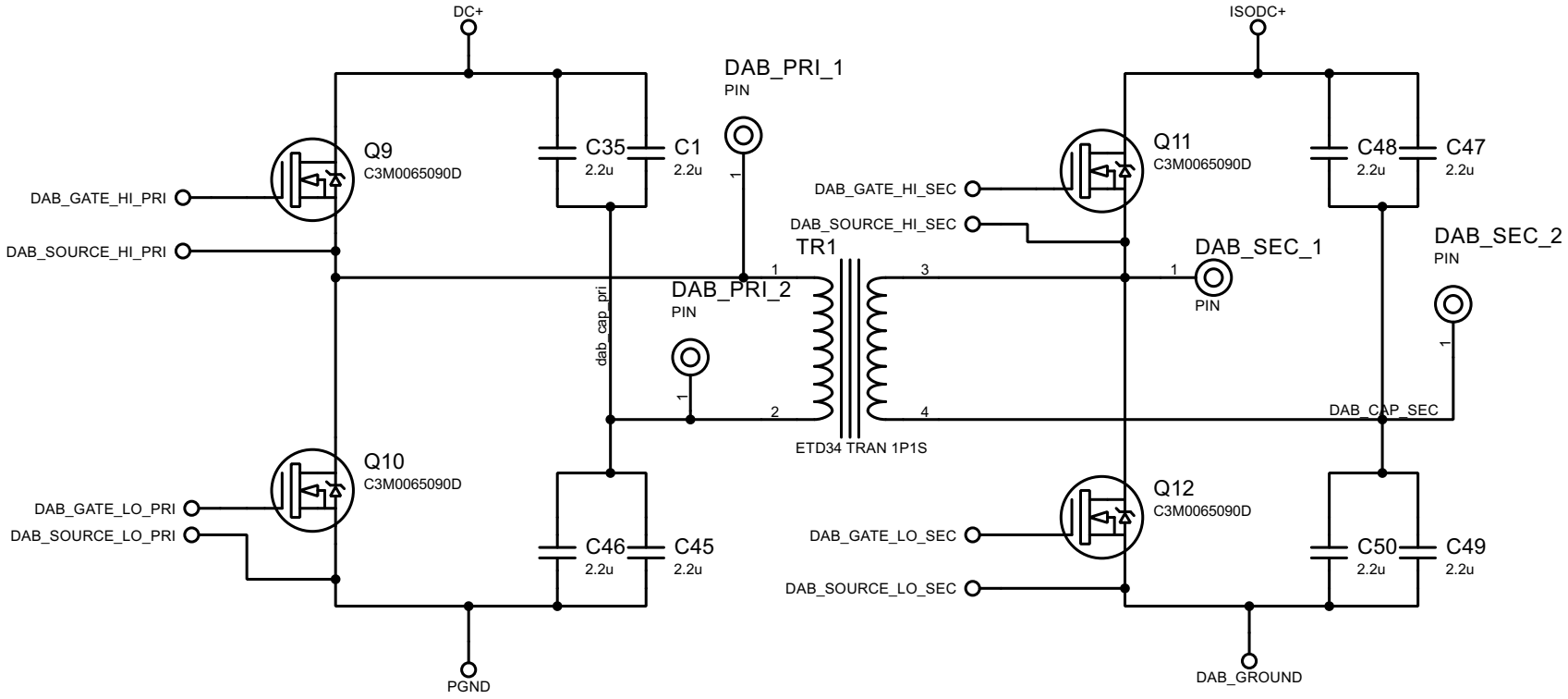


FILE NAME:	single_phase_inverter_psu_v3.pdsprj
DESIGN TITLE:	Inverter PSU
PATH:	regulators
BY:	JH
REV:	0
DATE:	26/01/2025
PAGE:	3 of 11
TIME:	11.34.56

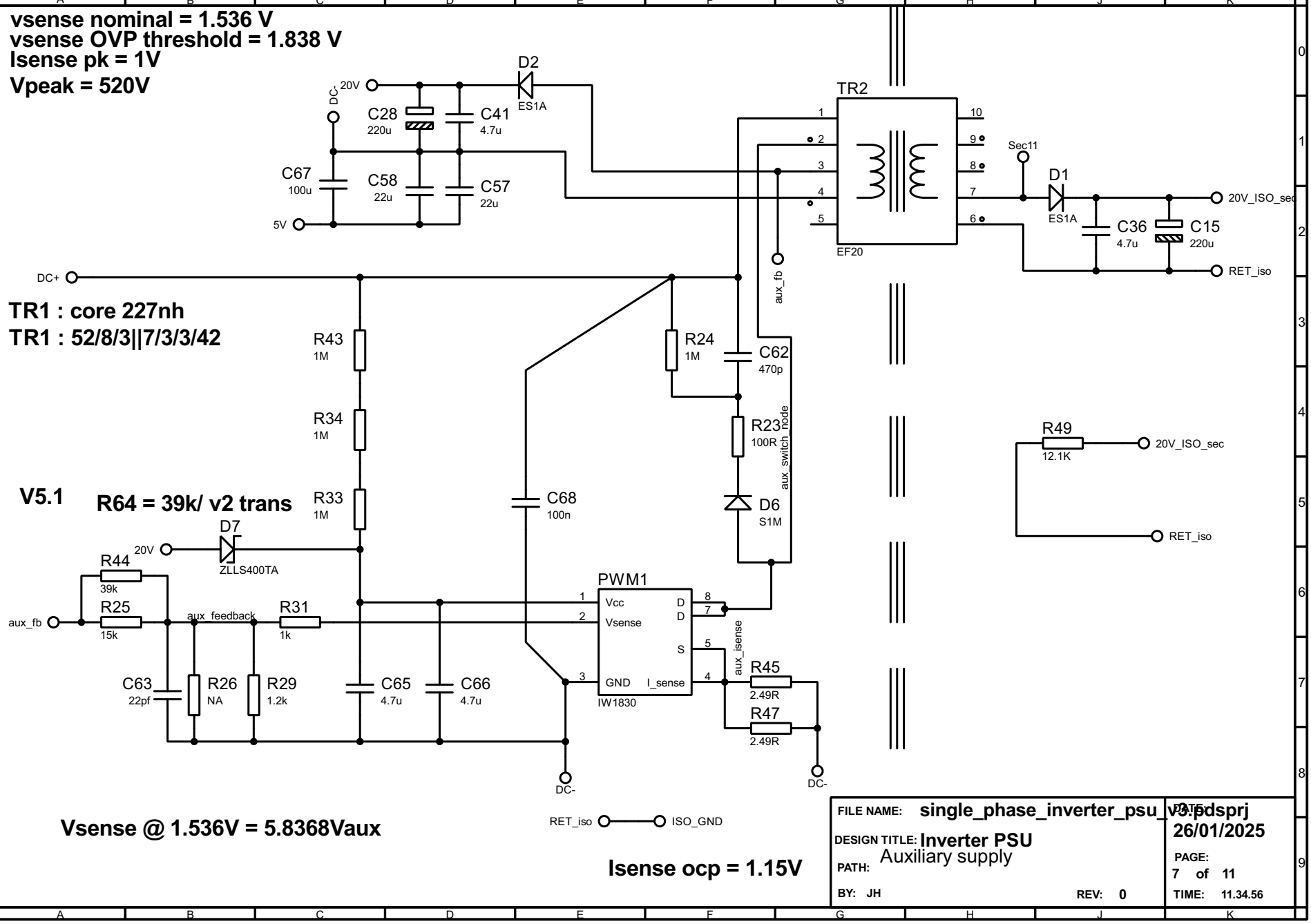


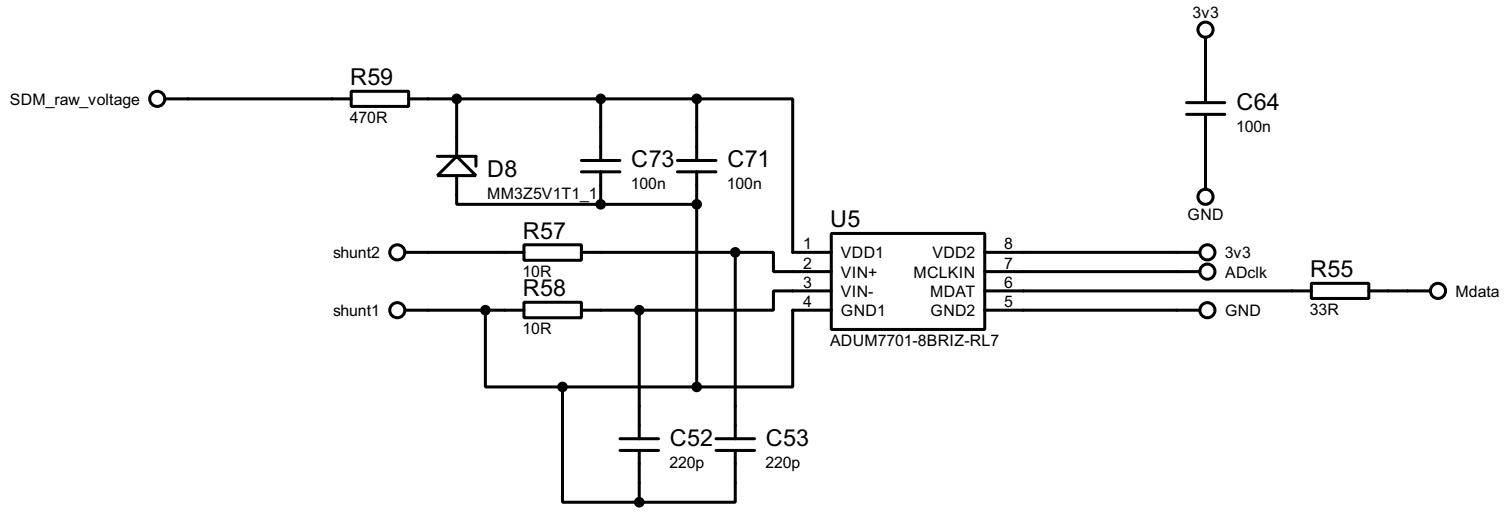
FILE NAME:	single_phase_inverter_psu_v3.pdsprj
DESIGN TITLE:	Inverter PSU
PATH:	DC links
BY: JH	REV: 0
DATE:	26/01/2025
PAGE:	4 of 11
TIME:	11.34.56

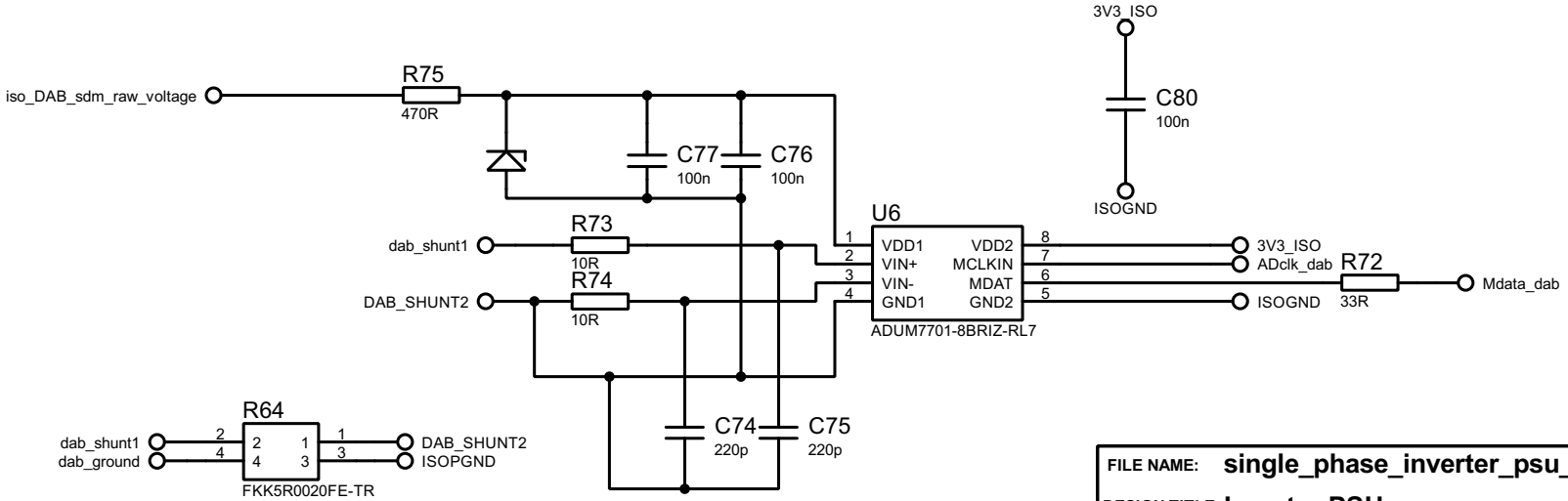

IKZ50N65EH5XKSA1
ISODC+
Q5
C3M0065090D
Q7
C3M0065090D
Q6
C3M0065090D
Q8
C3M0065090D
Q5_G
Q5_S
Q6_G
Q6_S
Q7_G
Q7_S
Q8_G
Q8_S
ISOPGND
INU_SHUNT1
INU_SHUNT2
R91
FKK5R0020FE-TR
inductor_input
INU_OUT2
PIN
J7
J9
J10
inductor_output
inu_Filter_input
20V_ISO_SEC
RELAY_DRIVE_ISO
R79
COIL_2
COM_2
NO
COIL_1
COM_1
SLA-12VDC-SL-A
afe_relay
AC_out_1
AC_out_2
ACOUT
ACOUT2
CM3
WE-CMBNC
C33
7u
C34
7u
C30
2n2
C29
2n2
C32
2n2
C31
2n2
PE6
PE7
R60
R62
R61
R63
R71
R256
91K
R65
R70
R66
R69
R67
R68
R46
10k
REF_1V65_ISO
3V3_ISO
U7:B
R56
8.2k
ISOADA2
OPA2353UA
ISOGND
R51
10k
R52
10k
U7:A
R54
8.2k
ISOADA1
R53
10k
C38
100n
C56
100n
PE5
PE6
PE7
FILE NAME: single_phase_inverter_psu_v3.pdsprj
DESIGN TITLE: Inverter PSU
VSI
PATH:
BY: JH
REV: 0
DATE: 26/01/2025
PAGE: 5 of 11
TIME: 11.34.56



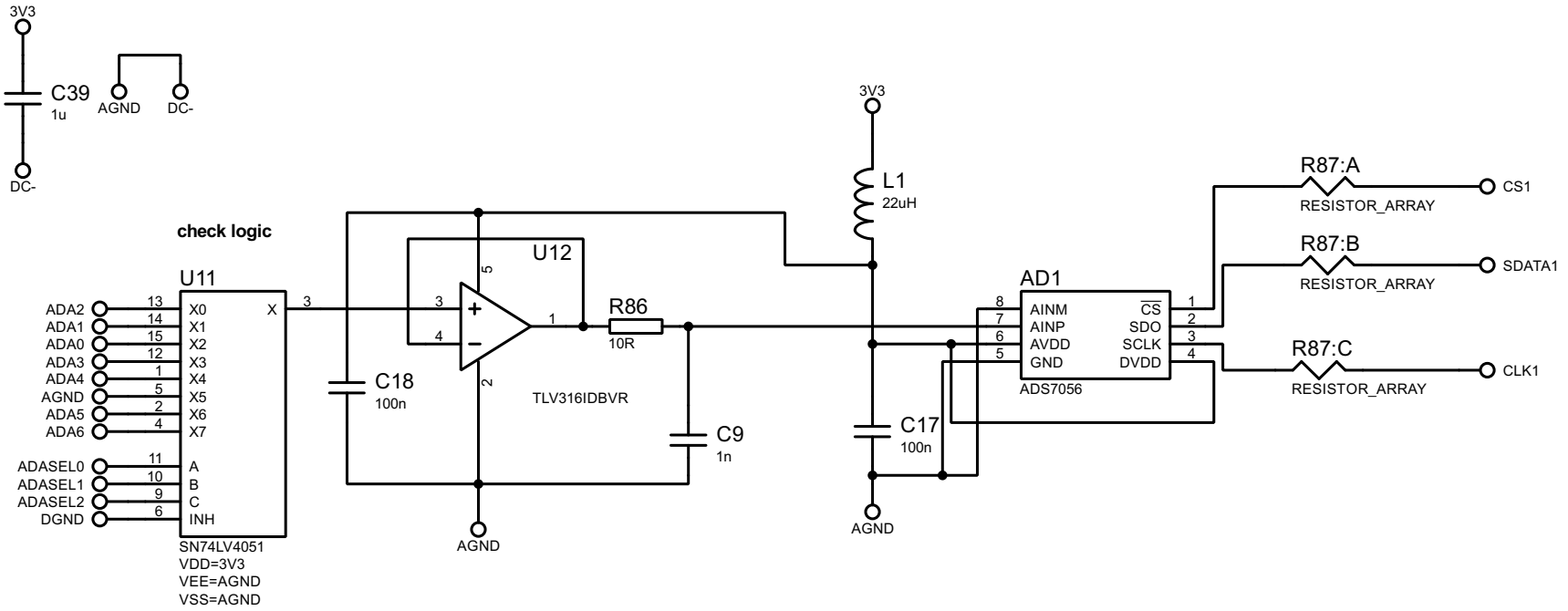
FILE NAME: single_phase_inverter_psu_v3.pdsprj	DATE: 26/01/2025
DESIGN TITLE: Inverter PSU Dual Active Bridge	PAGE: 6 of 11
PATH:	TIME: 11.34.56
BY: JH	REV: 0


vsense nominal = 1.536 V
vsense OVP threshold = 1.838 V
Isense pk = 1V
Vpeak = 520V
TR1 : core 227nh
TR1 : 52/8/3||7/3/3/42
V5.1
R64 = 39k/ v2 trans
Vsense @ 1.536V = 5.8368Vaux
Isense ocp = 1.15V
DC+
DC-
20V
5V
D2
ES1A
C28
220u
C41
4.7u
C67
100u
C58
22u
C57
22u
TR2
EF20
Sec11
D1
ES1A
C36
4.7u
C15
220u
20V_ISO_sec
RET_iso
aux_fb
R43
1M
R34
1M
R33
1M
R24
1M
C62
470p
R23
100R
aux_switch_node
D6
S1M
C68
100n
R49
12.1K
D7
ZLLS400TA
R44
39k
R25
15k
aux_feedback
R31
1k
C63
22pf
R26
NA
R29
1.2k
C65
4.7u
C66
4.7u
PWM1
Vcc
Vsense
GND
D
D
S
I_sense
IW1830
aux_isense
R45
2.49R
R47
2.49R
RET_iso
ISO_GND
FILE NAME: single_phase_inverter_psu_v3.pdsprj
DESIGN TITLE: Inverter PSU
Auxiliary supply
PATH:
BY: JH
REV: 0
DATE: 26/01/2025
PAGE: 7 of 11
TIME: 11.34.56

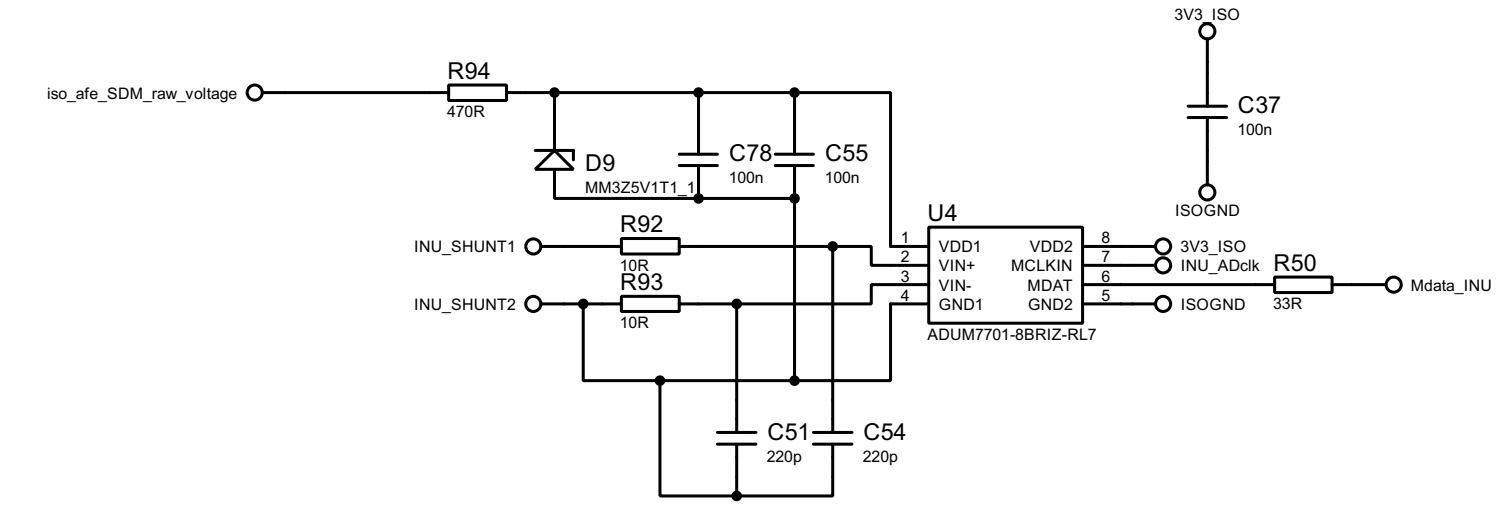


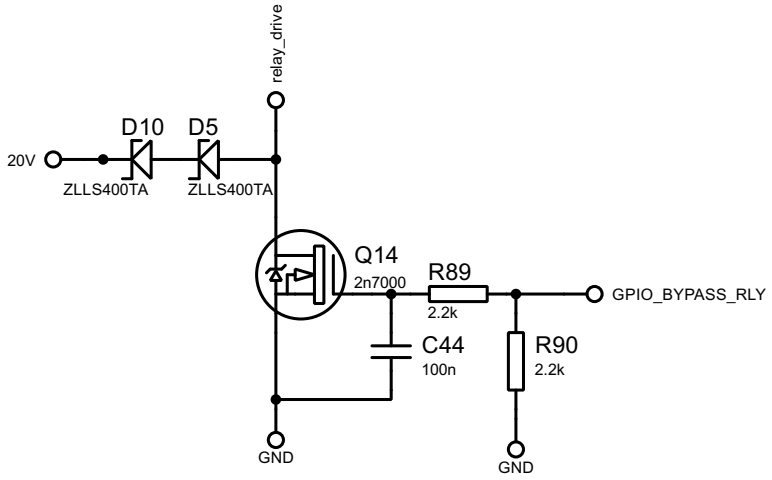


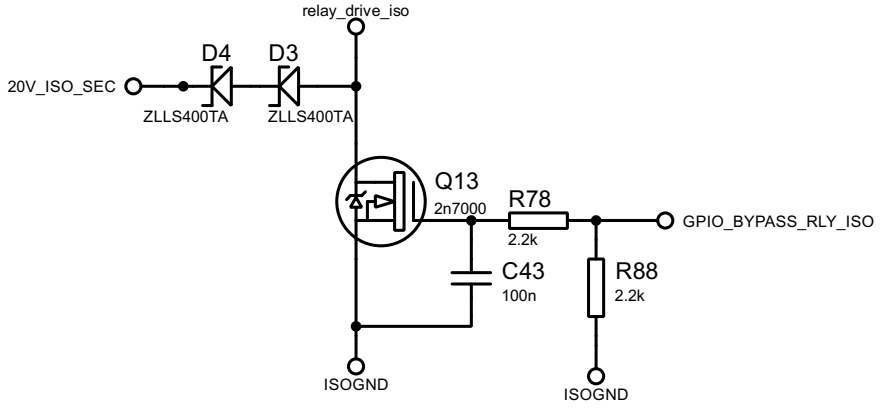
FILE NAME: single_phase_inverter_psu_v3.pdsprj	DATE: 26/01/2025
DESIGN TITLE: Inverter PSU SDM current measurements	PAGE: 8 of 11
PATH:	TIME: 11.34.56
BY: JH	REV: 0



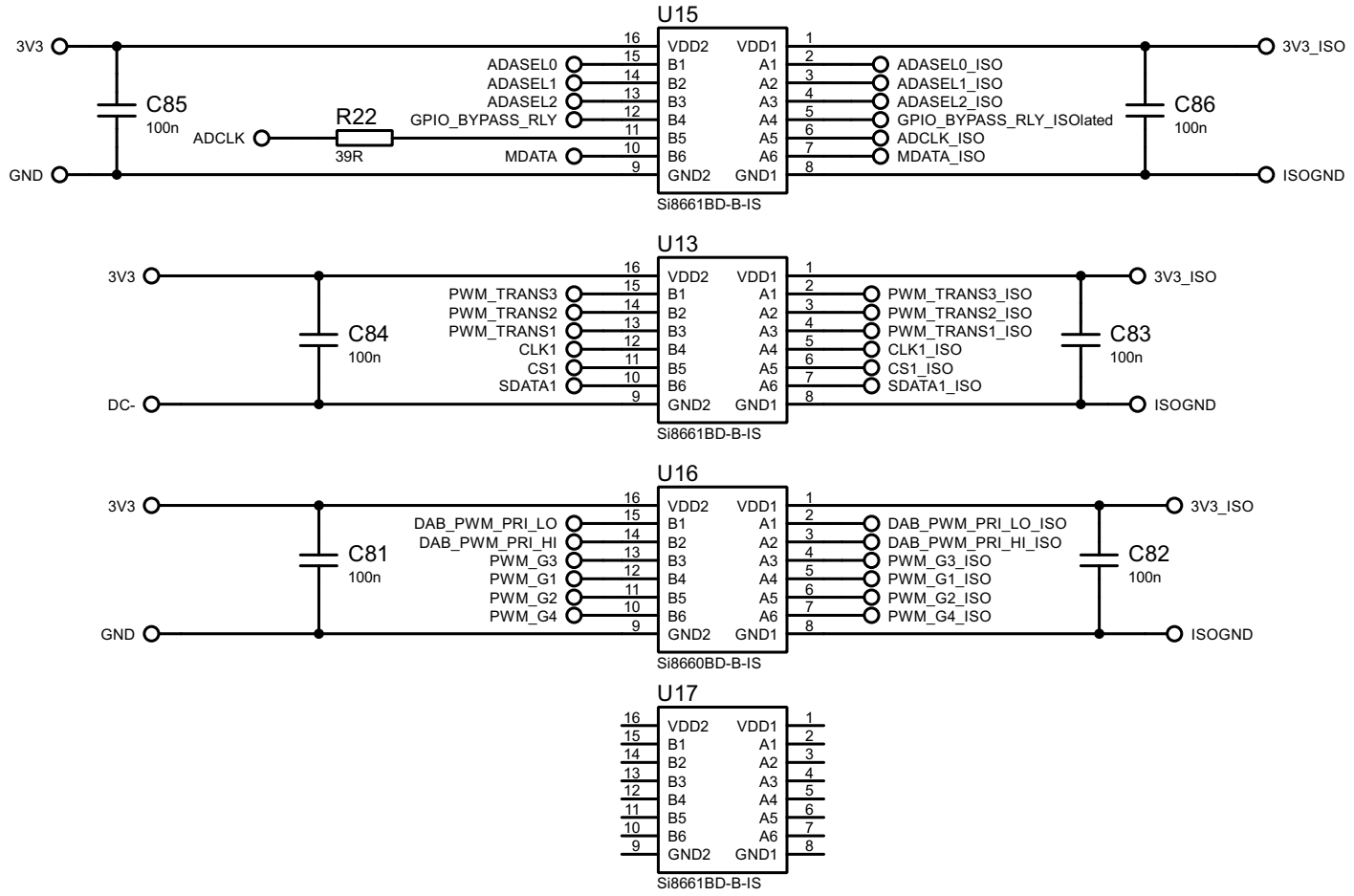
FILE NAME: single_phase_inverter_psu_v3.pdsprj	DATE: 26/01/2025
DESIGN TITLE: Inverter PSU primary side ADC	PAGE: 9 of 11
PATH:	TIME: 11.34.56
BY: JH	REV: 0







FILE NAME: single_phase_inverter_psu_v3.pdsprj	DATE: 26/01/2025
DESIGN TITLE: Inverter PSU	PAGE: 10 of 11
PATH:	TIME: 11.34.56
BY: JH REV: 0	



FILE NAME: single_phase_inverter_psu_v3.pdsprj	DATE: 26/01/2025
DESIGN TITLE: Inverter PSU signal isolators	PAGE: 11 of 11
PATH:	
BY: JH REV: 0	TIME: 11.34.56